

Supplementary Figures

Panel alignment is a precondition for cross-cancer causal comparison: the reproducible structure across 33 TCGA cohorts is a microenvironmental chemokine axis

Functional & Integrative Genomics

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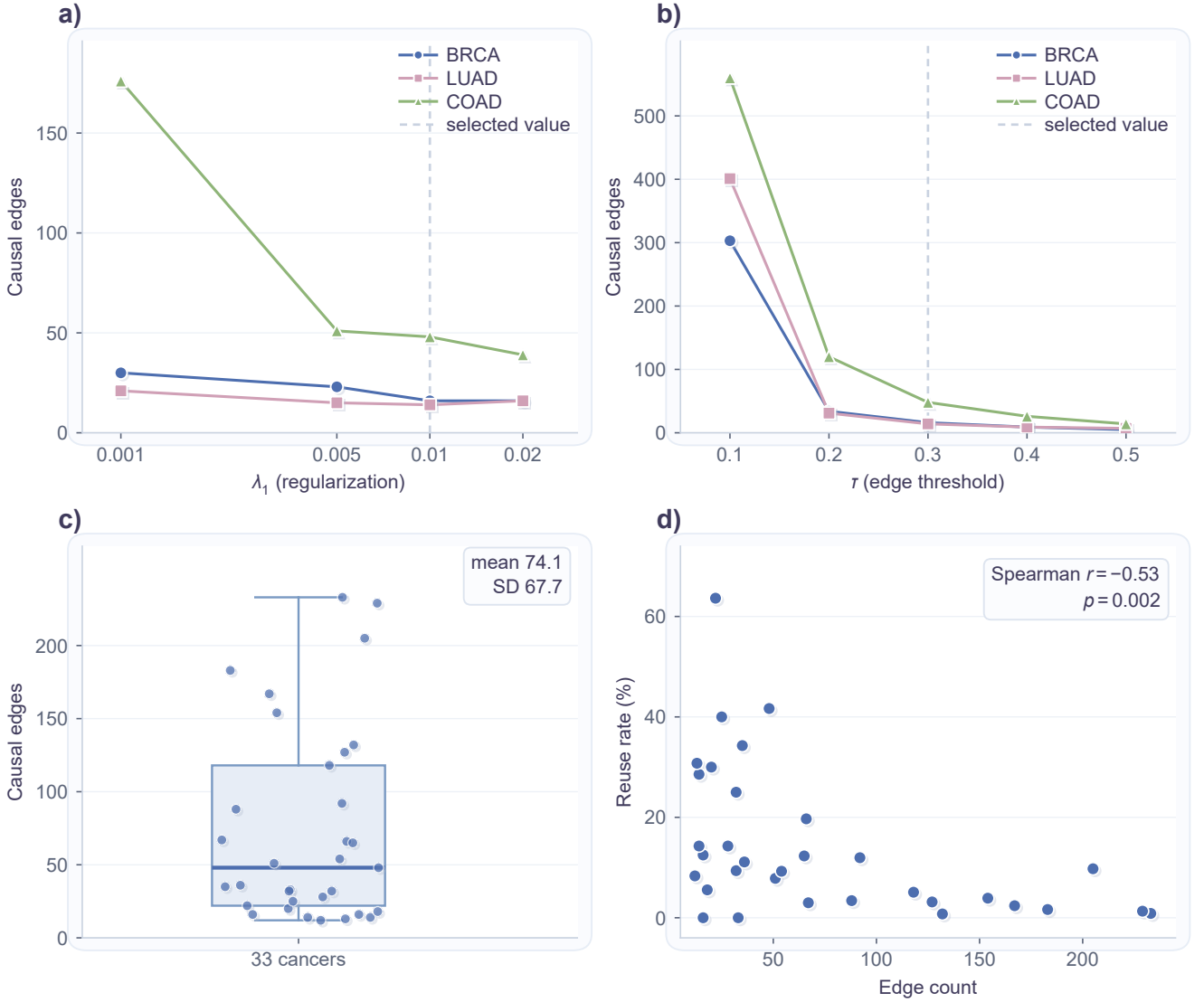


Fig. S1 Parameter sensitivity of the NOTEARS pipeline. **a)** ℓ_1 regularization sensitivity for three representative cancers; $\lambda_1 = 0.01$ selected (dashed). **b)** Edge threshold τ sensitivity; $\tau = 0.3$ selected (dashed). **c)** Edge count distribution across 33 cancers (boxplot; mean 74.1, SD 67.7). **d)** Edge count vs per-cancer reuse rate (Spearman $r = -0.53$, $p = 0.002$)

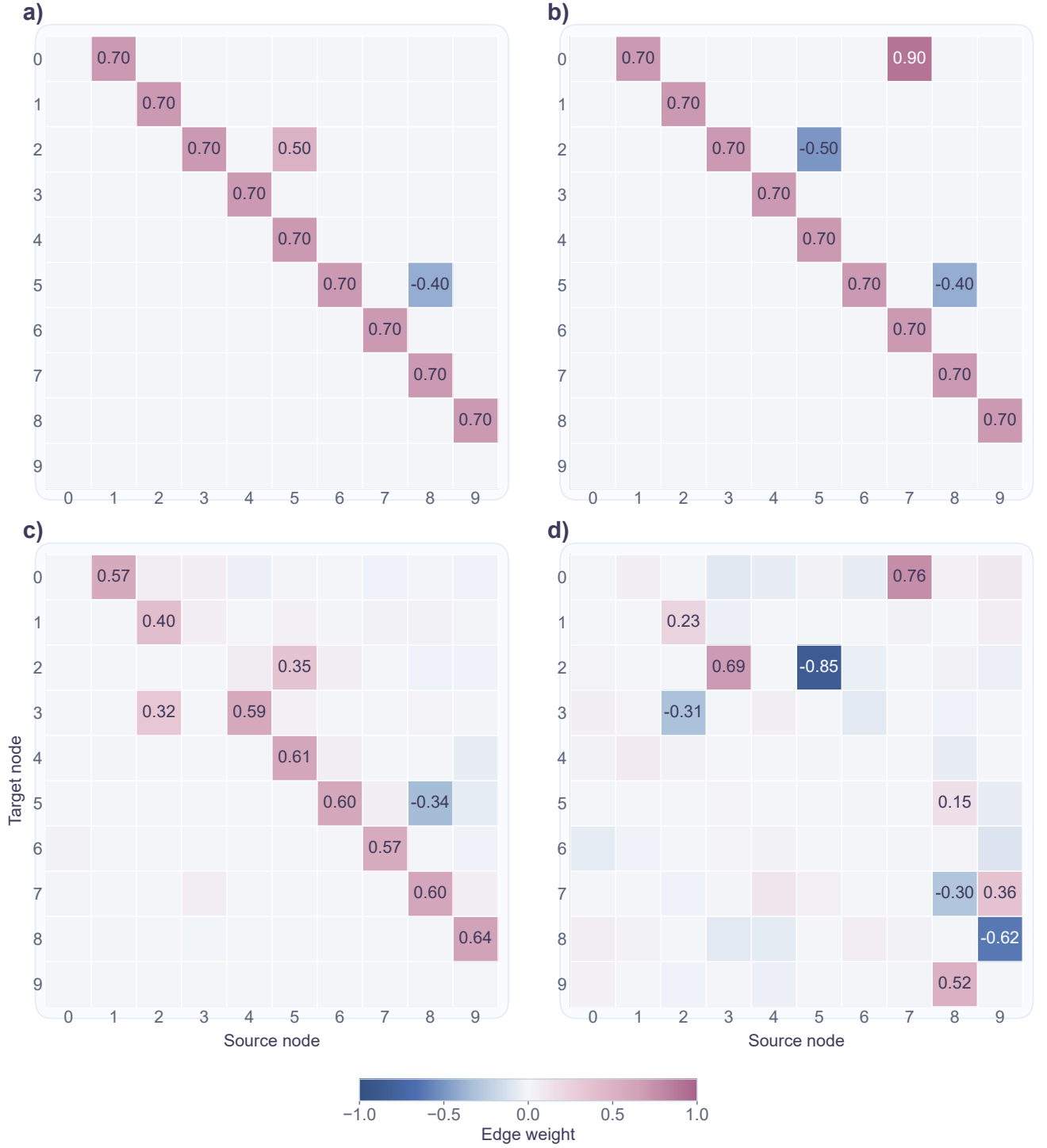
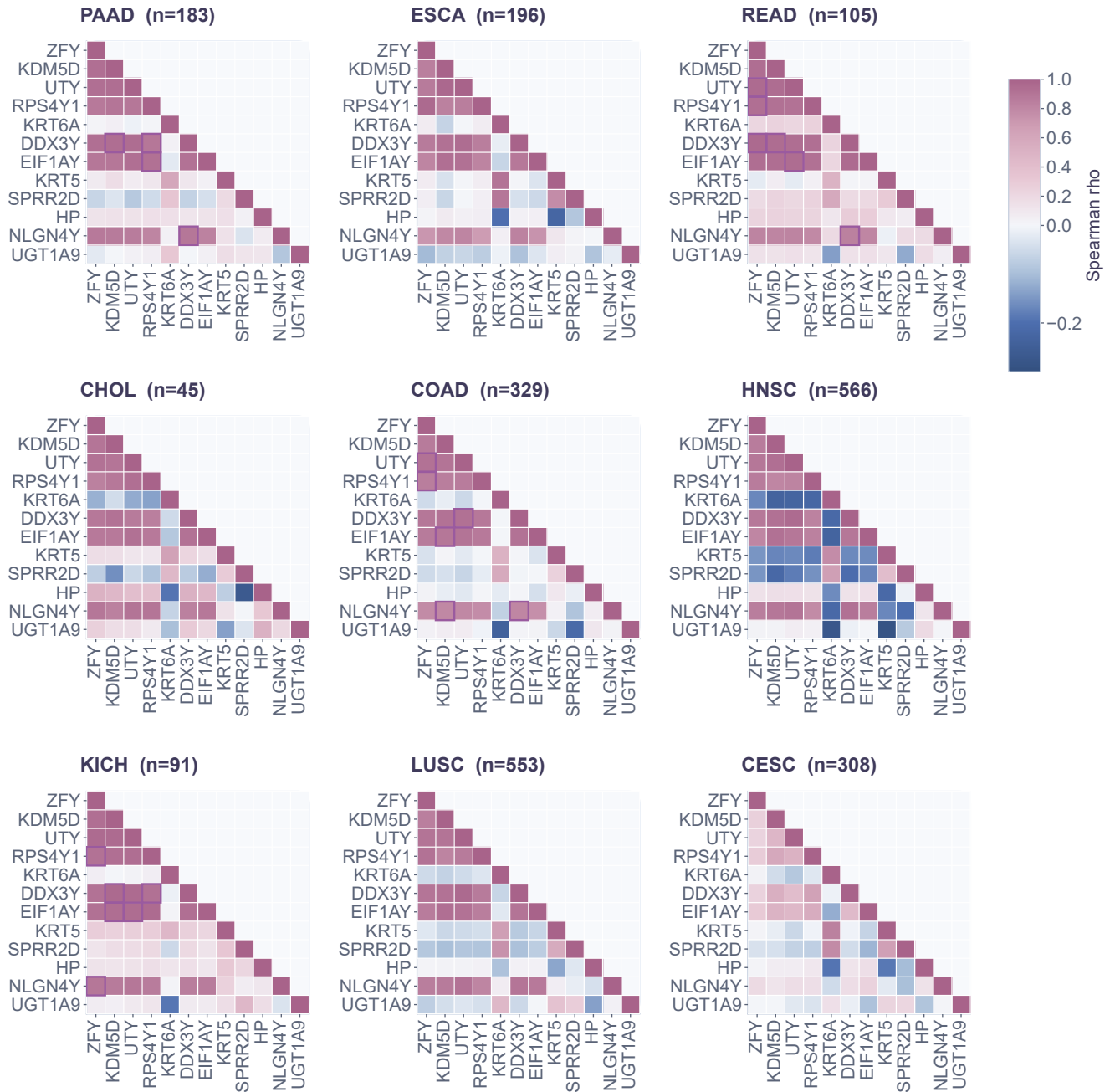


Fig. S2 Two-stage decomposition validation on a sparse chain-structured DAG ($d = 10$, $K = 3$). **a)** Ground truth shared backbone W_0 (11 edges). **b)** Ground truth Batch 1: rewired edge $2 \rightarrow 5$ (sign-flipped) and new edge $0 \rightarrow 7$. **c)** Recovered $\hat{W}_0$ from Stage 2 DAG projection (10/11 shared edges above $\tau = 0.2$, $h(W_0) = 5.21 \times 10^{-6}$). **d)** Deviation matrix $\Delta_1 = \hat{W}_1 - \hat{W}_0$ for the rewired batch. Overall recall: 14/15 ground-truth edges recovered ($= 0.93$); see Table 2 of the main text for Precision, F1, FDR, and SHD



Boxed cells: a causal edge of ours in that cancer type that STRING also supports (22 cells across the nine panels).

Genes are the 12 highest-degree members of the STRING-supported network; the nine panels carry the most supported edges.

Lower triangles only. Spearman rho within each cohort, on tumors complete for all 12 genes.

The Y-chromosome and the keratin genes each form a correlated block, and the two blocks are negatively correlated.

Mean off-diagonal rho for these twelve genes is +0.308; the cohort mean is positive in all 33 cancer types.

Fig. S3 Co-expression structure of the supported network. Lower triangles of the Spearman correlation matrix of the twelve highest-degree genes of the STRING-supported network, computed within each of the nine cohorts carrying the most supported edges. Boxed cells mark a causal edge of ours that STRING also supports. The Y-chromosome genes and the keratin genes each form a correlated block, and the two blocks are negatively correlated

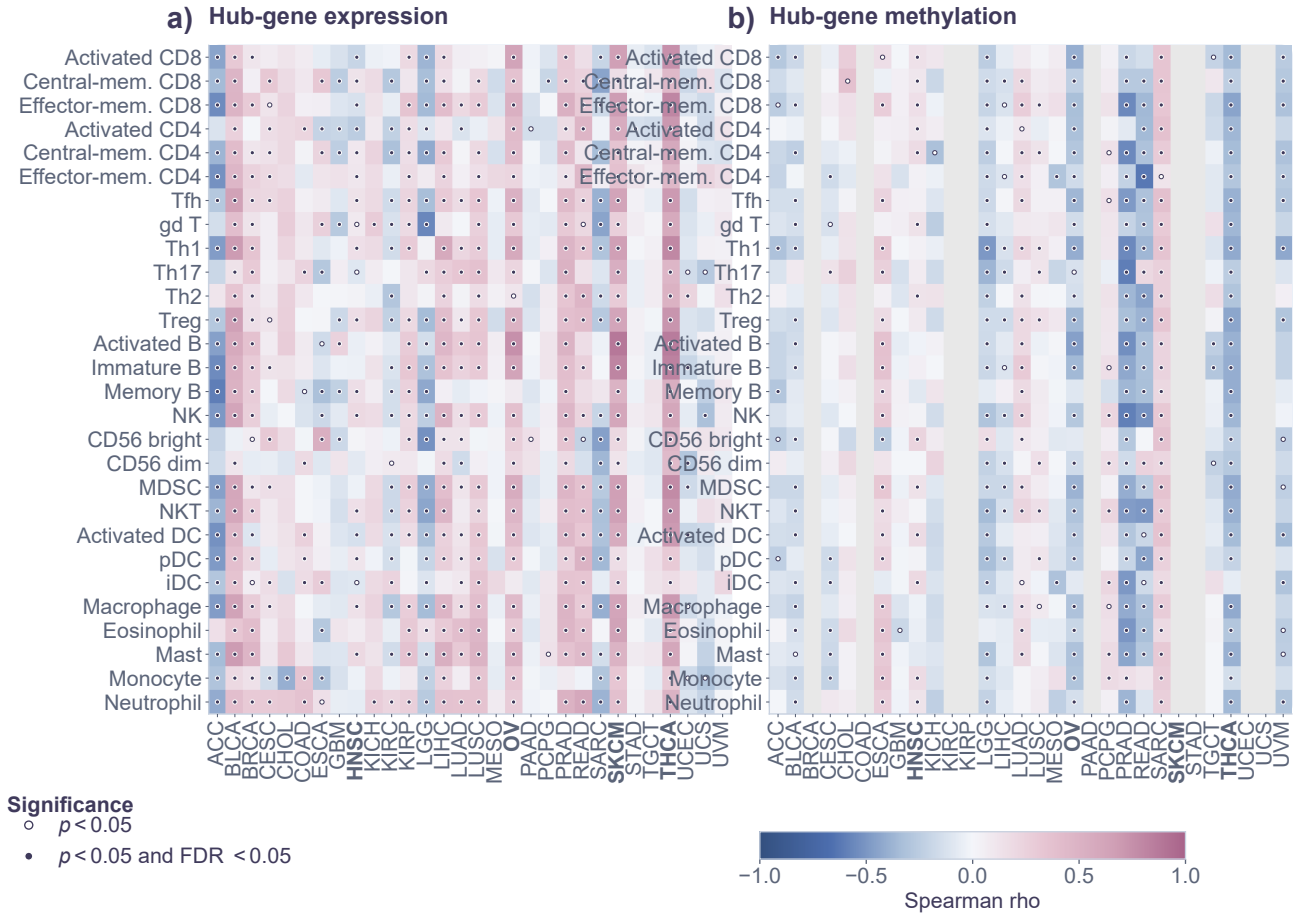


Fig. S4 Immune-microenvironment association of each cohort's own hub gene, with immune cell-type abundance taken from TISIDB (Ru et al., 2019). **a)** Hub-gene expression and **b)** hub-gene methylation against the abundance of the 28 TISIDB immune cell types, for the 30 cohorts carrying an immune profile; each cell is the Spearman ρ between that cohort's own hub gene and that cell type, with significance marked as in the legend. Bold cohort labels mark hubs that are themselves immune-lineage markers; gray columns carry no data, methylation covering 21 cohorts. Expression is predominantly positive (306 of the 416 significant pairs) and methylation predominantly negative (175 of 270); the copy-number layer, with only 44 of 756 pairs significant, is not drawn. Excluding the four immune-lineage hubs, the hub still exceeds the other genes of its own network in 57.7% of 728 pairs ($p = 4.4 \times 10^{-12}$), a modest effect (median $|\rho|$ 0.145 versus 0.122) that reverses in 10 of 26 cohorts